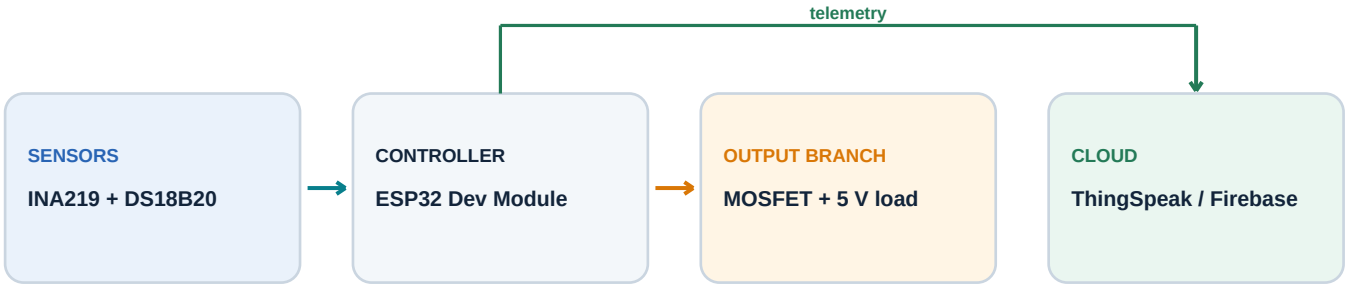
CHARGESENSE / STATION 01

ESP32 Hardware Connection Guide

Pin-by-pin wiring and low-voltage power-path figures for the firmware in this project.

SENSORS, OUTPUT, AND CLOUD RELATIONSHIPS



IMPORTANT: LOW-VOLTAGE PROTOTYPE ONLY

Use a current-limited, regulated 5 V supply and a known, rated demo load (for example, a small 5 V USB fan). **Do not connect mains AC, an unprotected battery, or a phone battery.** This prototype is not a certified charger or battery-protection system.

Actual ESP32 board and MOSFET module are unspecified. Verify board pins, module terminal map, logic compatibility, and ratings before powering the load.

WHAT YOU NEED

ESP32 Dev Module (or matching board) | INA219 current/voltage breakout | DS18B20 temperature sensor | 4.7 kohm resistor | rated 5 V MOSFET load-switch module | active buzzer module | LED + 220 ohm resistor | momentary pushbutton | current-limited 5 V supply | known 5 V demo load

01 / SIGNAL AND SENSOR WIRING

Connect modules to ESP32 GPIO

The board illustration is functional, not a physical pinout. Use the GPIO labels printed on your exact ESP32 board.

INA219 BREAKOUT

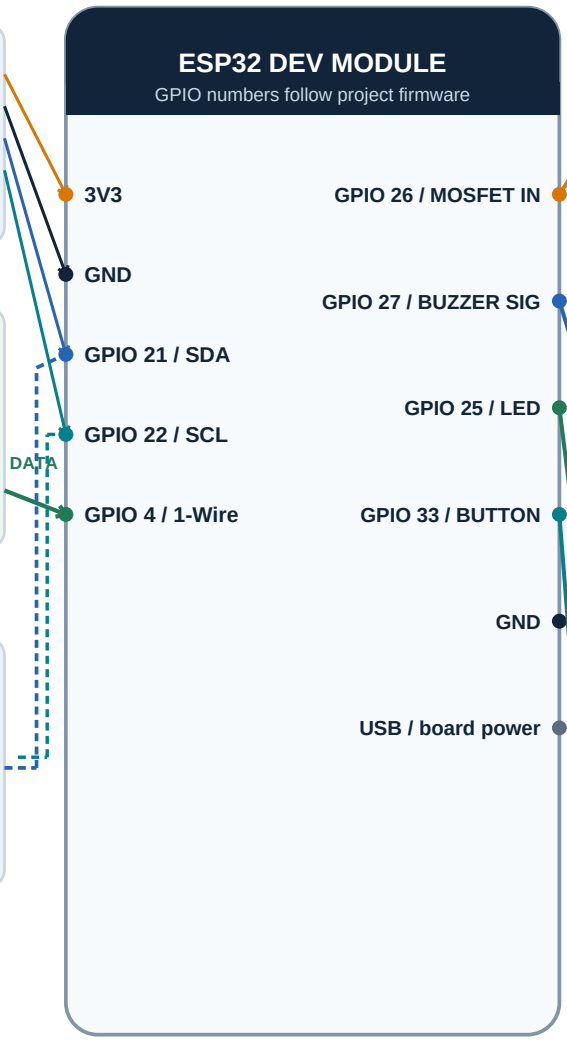
VCC -> 3V3
GND -> common GND
SDA -> GPIO 21
SCL -> GPIO 22

DS18B20

VDD -> 3V3
GND -> common GND
DATA -> GPIO 4
4.7 kohm: DATA to 3V3

OPTIONAL SSD1306 OLED

VCC -> 3V3 | GND -> GND
SDA -> GPIO 21
SCL -> GPIO 22
Shares the I2C bus



MOSFET CONTROL

IN / EN -> GPIO 26
GND -> common GND
VCC only as module requires
Verify 3.3 V logic + polarity

ACTIVE BUZZER MODULE

SIG -> GPIO 27
GND -> common GND
VCC per module rating
Use transistor if needed

STATUS LED

GPIO 25 -> 220 ohm -> anode
LED cathode -> GND
Observe LED polarity

RESET BUTTON

One side -> GPIO 33
Other side -> GND
Firmware uses INPUT_PULLUP

WIRING RULES

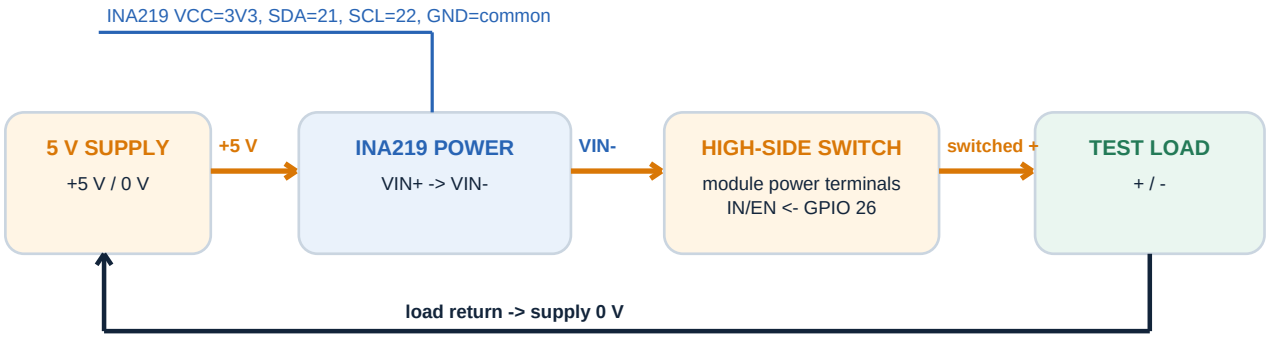
Tie ESP32 GND, sensor GND, module GND, and the low-voltage supply negative to a common reference where the selected MOSFET topology requires it. Keep GPIO logic at 3.3 V. Sensor breakout pin labels are authoritative; for a bare DS18B20, check the exact manufacturer datasheet rather than guessing lead order.

02 / CURRENT MEASUREMENT AND OUTPUT SWITCH

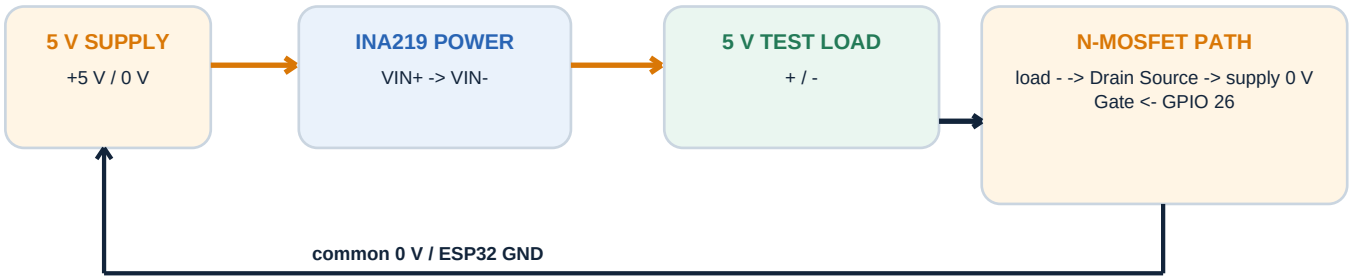
Wire the 5 V test-load path

The INA219 power terminals must be in series with the load current. Select only the switch arrangement that matches your exact MOSFET module.

A HIGH-SIDE SWITCH MODULE



B LOW-SIDE N-CHANNEL SWITCH (ONLY IF YOUR MODULE SUPPORTS THIS)



DO NOT GUESS THE MOSFET TERMINALS

The project does not identify the exact MOSFET board. The two figures are alternative power-path templates, not instructions to combine both. Follow the board manufacturer's schematic and terminal labels; verify its voltage/current/thermal ratings, 3.3 V control compatibility, and active polarity. Firmware assumes GPIO 26 is active HIGH by default. If your module is active LOW, its setting must be changed before use. If uncertain, leave the load disconnected.

03 / REFERENCE PIN MAP

Firmware GPIO connections

Pin mapping below matches the project's current ESP32 firmware. Verify your exact board silk-screen and module datasheets.

PART	MODULE PIN	CONNECT TO	NOTES
INA219	VCC / GND	ESP32 3V3 / GND	Logic supply; common ground.
INA219	SDA / SCL	GPIO 21 / GPIO 22	I2C bus; OLED can share it.
INA219	VIN+ / VIN-	5 V positive in series	Observe polarity; see page 3.
DS18B20	VDD / GND	ESP32 3V3 / GND	Use pin names, not assumed lead order.
DS18B20	DATA	GPIO 4	4.7 kohm pull-up from DATA to 3V3.
MOSFET module	IN / EN	GPIO 26	Check 3.3 V compatibility and polarity.
Active buzzer	SIG / GND	GPIO 27 / common GND	VCC per rating; transistor driver if needed.
Status LED	Anode / cathode	GPIO 25 via 220 ohm / GND	Do not omit series resistor.
Pushbutton	Two contacts	GPIO 33 / GND	Firmware uses INPUT_PULLUP.
Optional OLED	VCC / GND	3V3 / common GND	Use 3.3 V logic; share I2C pins.

OPTIONAL OLED

SSD1306 is disabled by default in firmware (`ENABLE_OLED 0`). If installed, it shares GPIO 21 (SDA) and GPIO 22 (SCL) with INA219. Use an OLED module with 3.3 V-compatible logic and install its libraries before enabling the feature.

04 / FIRST POWER-UP

Commission in a safe sequence

Do not attach the test load until each low-voltage signal and protection behavior has been checked.

1

POWER OFF

Disconnect the 5 V supply and remove the test load. Check the board GPIO labels, module datasheets, wire insulation, and the chosen MOSFET current-path topology.

2

CHECK SIGNALS

Wire INA219 logic to 3V3, GND, GPIO 21 and GPIO 22. Wire DS18B20 DATA to GPIO 4 and fit the 4.7 kohm pull-up from DATA to 3V3. Tie logic grounds together.

3

USB ONLY

Power the ESP32 from USB. Keep the 5 V load supply disconnected. Confirm no 5 V signal is connected to an ESP32 GPIO and verify the reset button is wired GPIO 33 to GND.

4

UPLOAD AND READ

Upload firmware and open Serial Monitor at 115200 baud. Confirm the INA219 and DS18B20 are detected. Firmware should keep the MOSFET output OFF at startup.

5

CHECK SWITCH

With no load attached, confirm the module is compatible with 3.3 V control and that GPIO 26 active-HIGH behavior actually leaves the output OFF at boot. Change firmware polarity only if the module requires it.

6

TEST LOW VOLTAGE

Use a current-limited 5 V supply and a known, rated demo load under supervision. Monitor voltage, current, and temperature. Stop immediately if readings are implausible or anything heats unexpectedly.

STARTER THRESHOLDS ARE NOT UNIVERSAL RATINGS

The firmware starts with 4.50-5.50 V limits, 1.00 A current maximum, a 40 C warning (clears below 38 C), and a 50 C cutoff. These are illustrative demo settings, not validated limits for your exact parts. Match protections to the supply, load, MOSFET, connectors, wire gauge, and sensor placement before energizing the load.

Never connect mains AC, an unprotected battery, or a phone battery to this prototype.